

Earth Science Bachelor Thesis

Adsorption of fluoranthene, fluorene, phenanthrene and pyrene
to Norit GAC830W
using passive dosing by silicone O-rings

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Index

1. Summary	1
2. Introduction	1
2.1. Black carbon adsorption research	1
2.2. Passive dosing	1
2.3. Relevance	1
3. Methods	2
3.1. Adsorption model	2
3.2. Experimental configuration	2
3.3. Passive dosing by O-rings	3
3.4. Adsorption	3
3.5. Recovery	3
3.6. Silicone : water distribution	4
3.7. HPLC-FLU analysis	4
4. Results	5
4.1. Mass balance	5
4.2. Distribution coefficients	8
4.3. Isotherms	8
4.4. Adsorption parameters	9
5. Discussion	10
5.1. Control sample set 6 variations	10
5.2. System losses	10
5.3. Adsorption parameters	10
5.4. Hypothetical isotherms	10
6. Conclusion	12
7. Sources	13

Annex 1 – Mass balance

Annex 2 – Distribution coefficients

1. Summary

A research into adsorption of 4 polycyclic aromatic hydrocarbons to active carbon is described, including a new passive dosing method, and its results presented. Adsorption is measured but results do not conform to literature values and are expected to be imprecise and inaccurate. Suggestions for method improvement are made including calculation of expected results; evaluation of the new passive dosing method is postponed until more accurate results are obtained.

2. Introduction

Black carbon adsorption research

Research in the field of black carbon adsorption is relatively new. The principle of adsorption of organic pollutants to organic matter in soils and sediments has become common knowledge since the late 1970's but was thought to be a linear process. A decade later, several studies started showing results implying multiple adsorption processes. Since the late 1990's, black carbon has been identified as having an adsorption capacity for certain compounds that is much higher than that of general soil and sediment organic matter (Koelmans et al., 2006).

Research into the specifics of black carbon adsorption has focused on adsorption to sediments containing black carbon (Cornelissen et al., 2006), the adsorption rates of pollutants to different types of black carbon (Jonker and Koelmans, 2002), pollutant adsorption to black carbon with well known properties (Amstaetter et al., 2012), several different theoretical models to explain the observed black carbon adsorption isotherms (Kleineidam, 2002), and mechanistic considerations such as planarity of pollutants and surface or pore sorption (Cornelissen et al., 2005). Knowledge on most of these topics is as of yet incomplete.

The research will concentrate on gathering information on the adsorption parameters of four PAHs to activated (black) carbon; over the past few decades, it has become clear that black carbon is probably the most important binding phase for PAHs in the natural environment (Koelmans et al., 2006). The parameters of adsorption will be studied, as well as the efficiency of a new experimental method. Research in this subject is hindered by two main factors.

First, the hydrophobicity of pollutants such as PAHs prevent the use of water solutions; maximum water concentrations are very low, attained slowly and deplete rapidly. Adding solvent with dissolved PAHs can affect the black carbon and is an impractical method due to limits posed by the cosolvent effect that changes the properties of the water, solvent mixture if it consists of >1% solvent (Schwarzenbach et al., 2003).

Second, the presence of tiny black carbon particles in the water can have strong influence in the measurement of dissolved PAHs, more so since they have more readily accessible surface areas and a faster adsorption rate (Yang and Al-Duri, 2005) and water concentrations are so low. Separating the particles is a very time- and effort-intensive process.

Passive dosing

Two papers in toxicology (Smith et al., 2010, and Smith et al., 2010) detailed a method of passive dosing by silicon elements. The application of this method to research in black carbon adsorption is, to the knowledge of the author, new. The silicon elements used can also help avoid the difficulties involved in measuring water PAH content by inducing water concentrations from the change in PAH concentration in the silicon elements combined with a known silicon:water distribution coefficient, and by extrapolating black carbon adsorption parameters from the same data under the assumption that all PAHs are present in either the silicon, the water or the black carbon, a method similar to the POM-SPE method of Jonker (2001).

Relevance

As black carbon particles have different global cycles and rates of uptake by organisms, transport

and deposition and bioavailability of PAHs are affected by adsorption to BC. Understanding the details of the adsorption process is important in assessing the effective threat of environmental PAHs to ecosystems and human health, since adsorbed PAHs have different mobility and bioavailability than freely dissolved PAHs. It also influences the possibilities of using BC for remediation of the harmful effects of pockets of high PAH concentration.

This has increased social relevance as the European Council has issued the Water Framework Directive, stating the need for all member states to improve the chemical and ecological status of water bodies by 2015. The priority substances list of the WFD (EC, 2008) includes eight PAHs. Knowledge of PAH sorption to black carbon could help in assessing risks posed by PAH presence and effective bioavailability. Not only will this reduce exposure to the PAHs, it may also lead to a reevaluation of WFD safety thresholds for PAH presence. This can help free resources to remediate the presence of other contaminants, since lower thresholds may imply certain areas have already attained a sufficiently safe chemical status.

3. Methods and techniques

Adsorption model

Adsorption is a process that can be described by several models. Three are most common and are illustrated in figure 3.1: the linear, Freundlich and Langmuir models. The linear model holds that the ratio between water-phase and adsorbed concentrations is described by a single distribution coefficient, as illustrated by the straight isotherm. The Freundlich equation adds an exponent to represent lower adsorption rates at higher concentrations due to reduced site availability as illustrated by the curve of decreasing adsorption, and the Langmuir model introduces a maximum concentration beyond which the sorbent does not adsorb any additional sorbate, illustrated by the curve that approaches a maximum of adsorbed sorbate.

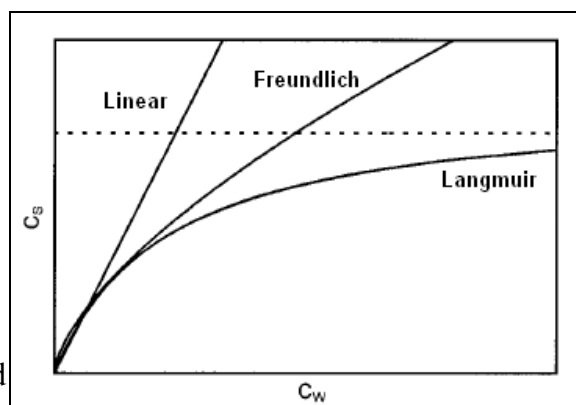


Figure 3.1: Linear, Freundlich and Langmuir adsorption models as described by their isotherm, or lines of related water-phase (C_w) and adsorbed solid-phase (C_s) sorbate concentrations.

The adsorption of PAHs to black carbon has been studied and its parameters extrapolated to form adsorption isotherms that were expected to fit a model from figure 3.1. Since Norit granulated active carbon 830W was used, black carbon will from now on be referred to as active carbon, or AC. It has properties very similar to black carbon, with a difference that it has been thermally treated to increase adsorption efficiency.

Experimental configuration

In order to simulate PAH adsorption to active carbon, erlenmeyers were prepared in triplicate containing active carbon and PAHs in water. These erlenmeyers were assumed to provide representative conditions during a seven day shaking period to allow adsorption to take place and provide representative data for the process in a natural environment.

An equilibrium state between the water and adsorbed PAHs was assumed to be attained within several days, and by analyzing the PAH contents of the carbon and the water, the details of adsorption were to be determined.

Analyzing the PAH concentrations in water is difficult since they are very low, and since difficult to remove small carbon particles will be present in the water which at such low concentrations can have a major effect on measured levels. Since water PAH concentrations are determined by the

loading ring PAH content and a distribution coefficient, they can also be induced instead of analyzed. The rings partially deplete during the experiment because the PAHs adsorb to the AC and are removed from the rings, so to induce water levels, the silicone:water distribution parameters and the O-ring PAH content after the adsorption experiment needed to be analyzed.

For most analyses samples, an Internal Standard was added; in the case of the loading solutions this IS was not added, an oversight. The IS consisted of anthracene in methanol, and was used in fixed quantities to determine any PAH losses during laboratory operations such as pipetting and distillation. Any loss in IS was assumed to be identical to loss in the PAHs present in the samples.

Passive dosing by O-rings

The concentration of PAHs was maintained through passive dosing. Silicone O-rings were pre-charged with PAHs and added to the adsorption samples, where they continuously released their PAH content. The charging was performed by preparing a single stock solution of 1000mg/L of all the four PAHs, after which six erlenmeyers were filled with decreasing strengths of this stock and up to 30 rings were added to them as detailed in table 3.1. The erlenmeyers were then allowed to shake in dark conditions for six days. After this period the rings were assumed to be charged with PAHs; the charging solutions and several of the rings were analyzed after loading to determine their exact PAH concentrations. For analysis, the charging solutions were diluted to reach PAH concentrations within the HPLC-FLU measurement range; the rings were prepared for analysis as described in the paragraph on recovery.

Charging solution	1	2	3	4	5
Initial PAH concentration (mg/L)	10	30	100	300	1000
Amount of rings added	15	15	15	15	30

Table 3.1: charging solutions for the silicone O-rings

Adsorption

Seven sets of samples were prepared, containing increasing levels of PAH concentrations, and every set consisted of 3 identical samples. They were prepared by creating a water stock of milliQ pure water, with 0,01M CaCl₂ for ion strength and 0,001M NaN₃ to reduce biological activity, of which approximately 75 mLs was added to each sample erlenmeyer. To each of these, 20 milligrams of AC and a set of 2 O-rings were added. Five sets were made of three erlenmeyers with O-rings from the five different charged concentrations; one control set (set 6) was made that did not contain any carbon, and one control set (set 7) did not contain any O-rings. Table 3.2 details the contents of each single erlenmeyer, per sample set.

Sample set	1	2	3	4	5	6	7
PAH conc. (µg/L)	10	30	100	300	1000	1000	0
Water (mL)	75 mL	75	75	75	75	75	75
Silicone rings	2	2	2	2	2	2	2
GAC830W (mg)	20 mg	20	20	20	20	20	20

Table 3.2: Adsorption samples

Recovery

PAH concentrations of the O-rings were removed to be analyzed with an Accelerated Solvent Extractor. This instrument uses high pressure and high temperature to quickly remove soluble fractions from solid phases with a choice of solvents. The rings used in the adsorption experiment were extracted one by one, while the rings that were analyzed to determine PAH levels before the adsorption experiment were extracted two at a time. Acetonitrile was used as a solvent and all extractions were performed twice to fully remove all PAHs; the method had been developed and tested for efficiency, and the rings had been analyzed for properties previously by E. Troia, master

student at the UvA.

PAH concentrations of the active carbon were extracted for analysis using pentane. The carbon was filtered from the adsorption samples, after which up to three extractions were performed where 5 mLs of pentane were added to the carbon and allow to shake for an hour. The pentane was subsequently removed and the process repeated with 2,5 mLs; the accumulated pentane was then spiked with a determined amount of IS, and distilled from around 10 mLs to 0,1 mL O-heptanol to reach concentrations measurable by the HPLC-FLU.

The same method was used for the extraction of PAH concentrations from the water in control sample set 6; this extraction was not hindered by the presence of small carbon particles since no carbon was added to set 6.

Silicone:water distribution

To ascertain the distribution coefficient between the O-rings and the water, the measurements of the recovered PAH concentrations from the O-rings and from the water phase of sample set 6 were used. The value was compared to literature values, assumed to be linear in relation to water PAH concentrations, and used to calculate the water concentrations in the other samples.

HPLC-FLU analysis

The analysis of the ASE-produced acetonitril solutions containing the PAHs from the O-rings that had been used in the adsorption experiment, the O-heptanol solutions containing the PAHs extracted from the carbon, and the O-heptanol solutions containing the PAHs extracted from the water from sample set 6, were transferred to HPLC vials for HPLC-FLU analysis.

The HPLC-FLU provides data output in the form of peak area response: a cumulative collection of fluorescence measurements per compound. In order to quantify these area responses to a concentration for the compound, the area responses of the HPLC-FLU needed to be calibrated. To this purpose, calibration levels were prepared of 8 known concentration of the four PAHs used in the research, and to them an internal standard was added as detailed in table 3.3 below. Level 8 was added later as an additional concentration and therefore does not follow the sequence of decreasing concentrations.

Sample (Calibration Level)	CL1	CL2	CL3	CL4	CL5	CL6	CL7	CL8
PAH concentration (µg/L)	500	200	100	50	10	5	2,5	20
IS concentration (µg/L)	100	100	100	100	100	100	100	100

Table 3.3: HPLC calibration levels

These levels provided typical peak area outputs along a concentration gradient, or calibration curve, for each used PAH. The peak area response from measurements could be compared to this gradient for quantification. However, since all samples were to be spiked with a known IS concentration, the relation between IS peak area and concentration could furthermore be used to circumvent the errors in measurement associated with losses from laboratory operations; this IS correction was included in quantification. The resulting method is represented by formula 3.1.

$$\frac{A_{PAH}}{A_{IS}} = a \left(\frac{C_{PAH}}{C_{IS}} \right) + b$$

Formula 3.1: quantification model for PAH measurements. A_{PAH} and A_{IS} are peak area responses for PAH and IS for the measurement quantified; C_{PAH} and C_{IS} are concentrations of PAHs and IS in the measurement.

The a and b in formula 3.1 are the parameters of the calibration curve and can be found by fitting a linear equation ($f(x)=ax+b$) to it. To obtain functional parameters, it was plotted in the form of the ratio between area responses for the PAHs and the IS, to the ratio between concentrations for the PAHs and IS; the same form as formula 3.1. An example of a calibration curve and the fitted

equation with resulting a and b for use in formula 3.1 can be seen in figure 3.1.

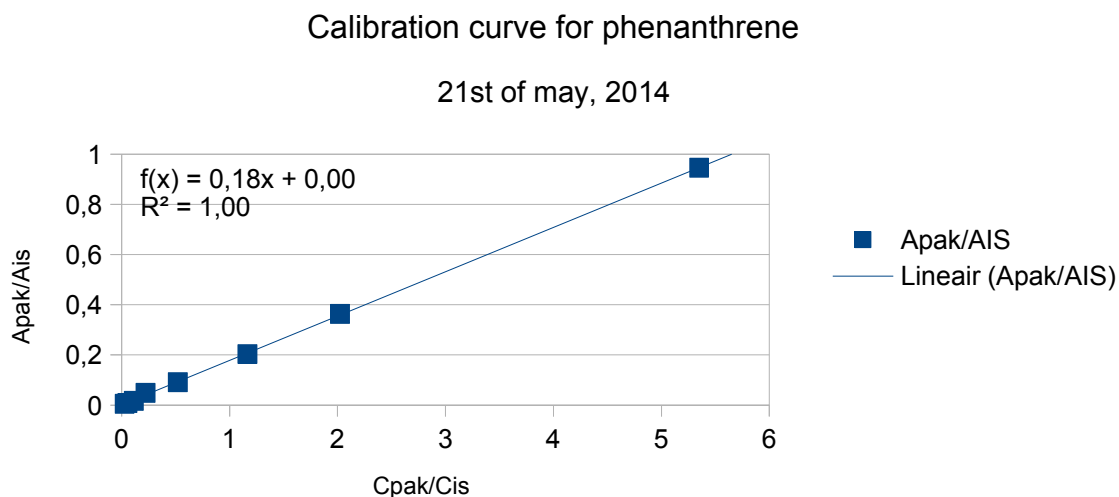


Figure 3.1: calibration curve, including linear fit equation.

The quantification formula was used to quantify measurements of HPLC-FLU analyzed PAH concentrations from all samples. The resulting measurement data was processed with Excel.

4. Results

First, a mass balance for the PAHs in all measured phases in the experiment can be seen in Annex 1. It contains the measurement data for the PAH contents of the charged rings, the rings after the adsorption experiment, the PAHs recovered from the carbon, calculations for the PAH contents of the water, and various deduced concentrations. The information has been summarized in figures 4.1 to 4.3.

Please note that water concentrations have been calculated based on a distribution coefficient derived from literature since measurements of the PAH contents of the water from control sample set 6, samples without any carbon, and the silicone:water K derived from them, have very low precision and are likely inaccurate. This is due to inefficient pentane extraction and distillation, the lack of correction with internal standard due to oversights in data notation and the calculation of required distillation factors. Also, carbon calculations have been made under the assumption that all active carbon has been recovered from the samples; and the carbon that has been recovered has not been completely dried due to time restrictions and overestimations in weight might be as large as 20%. Last, in extracting the PAHs from the carbon, all B samples have been triple extracted, all C samples have been double extracted, and all A samples have been double extracted and have had a third extraction from the C samples added to them due the authors lack of laboratory proficiency. This last oversight does not show a consistent effect on the amounts of PAHs extracted from the A, B and C samples.

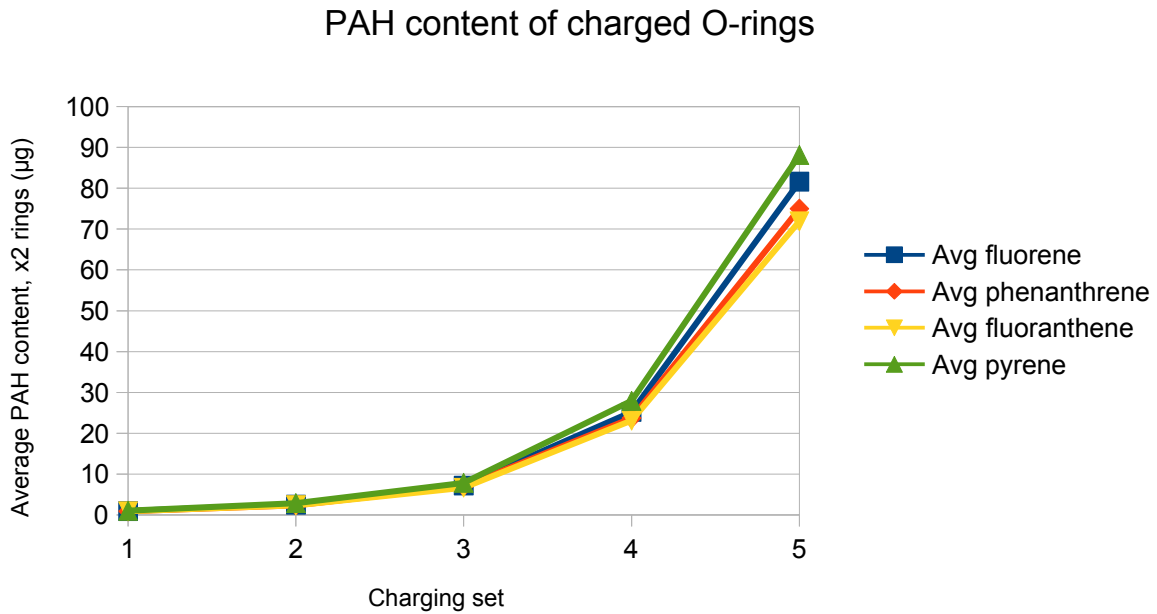


Figure 4.1: initial PAH contents of the charged O-rings, per charging set.

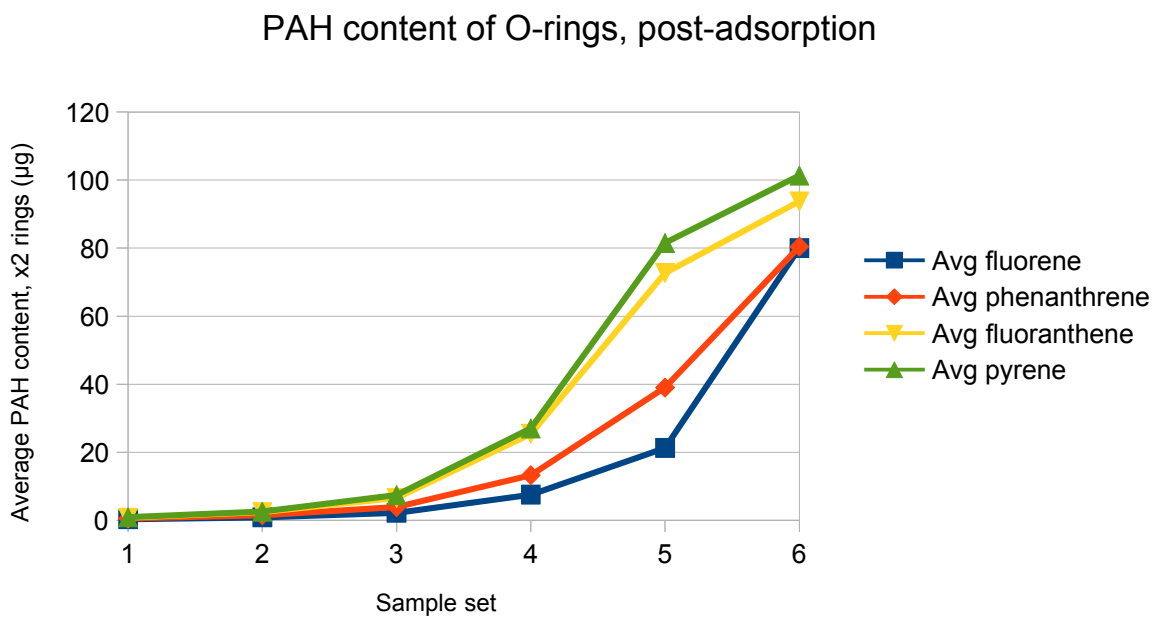


Figure 4.2: initial PAH contents of the post-adsorption O-rings, per sample set.

The first figure shows the initial PAH content of the O-rings, while figure 4.2 shows their content after the adsorption experiment. Note that the sample sets numbered 1 through 5 were prepared with rings from the charging sets of the same number and that for sample set 6, rings from charging set 5 were used; also, the rings were not identical – tested charged rings from charging sets were different unique rings as the ones used for and measured after the adsorption experiment. Measurements deviated from the average by around 5% for the charged rings, and up to 25% for the post-adsorption rings sets 1 through 5 while post-adsorption control set 6 showed heavy variation of up to 75%.

Since the water concentrations were calculated from the ring concentrations, a graph for water concentrations would look identical to figure 4.2 but with lower quantities ranging from 0,05 up to 6 µg.

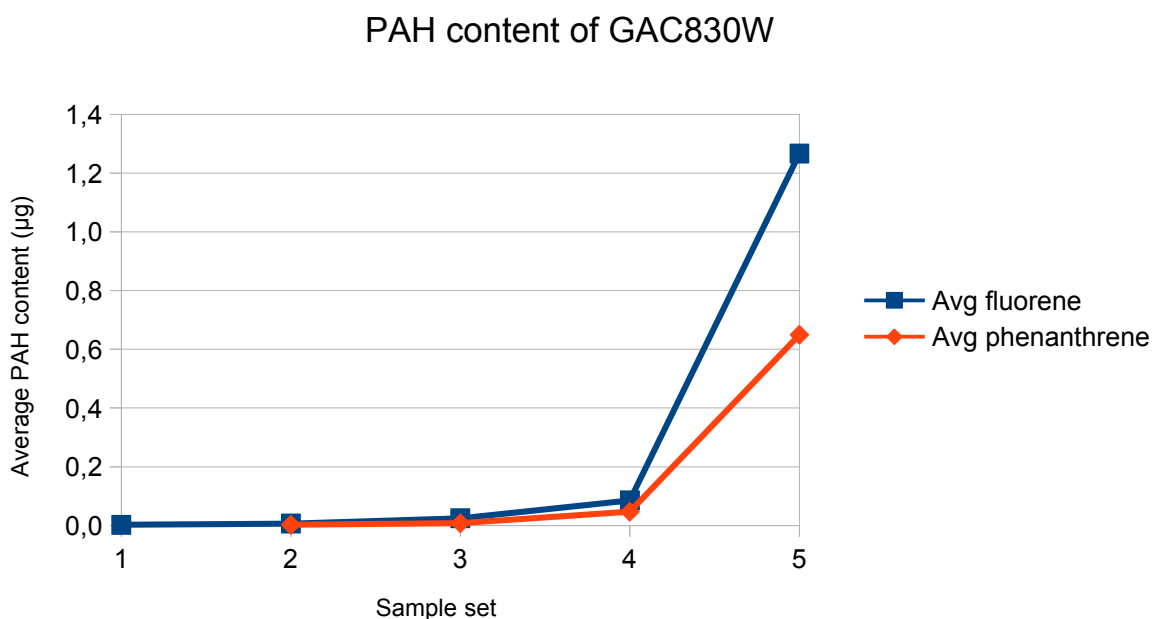


Figure 4.3: recovered PAH contents of active carbon.

Figure 4.3 shows PAH quantities extracted from the active carbon. Only two PAHs were measured in the extractions, fluorene and phenanthrene; variation reached up to 50% from average.

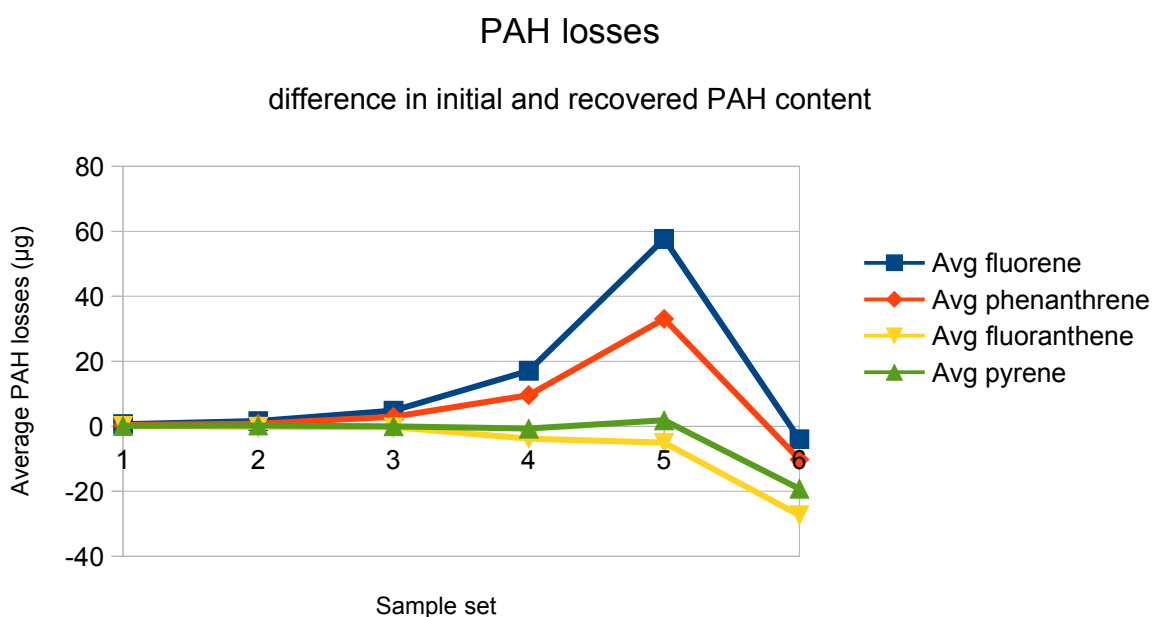


Figure 4.3: PAH losses: difference in initial PAH contents, and the sum of quantities extracted from post-adsorption rings, carbon and calculated water quantities.

Figure 4.4 shows the difference between the PAH contents of the charged rings, and the sum of recovered quantities from the post-adsorption rings, carbon and water; losses increase with concentrations. Losses were calculated from averaged values of charged and post-adsorption rings since the rings measured were not identical; high variation in post-adsorption set 6 is smoothed out in this graph. Note the negative losses in set 6.

Since the PAH contents extracted from the carbon and the water were relatively low compared to the ring contents, the loss graphs ia very similar to the dosing graph showing the amount of PAHs

released by the rings.

Table 4.1 below summarizes some of the findings in numbers; it shows average initial and post-adsorption PAH contents for the rings, water, and carbon, and system losses.

Sample set (average of three)	1	2	3	4	5	6	7
Initial ring PAH content, measured, mg/L	2,563	5,741	19,42	67,67	214,4	214,4	N/A
Post-adsorption ring PAH content, measured, mg/L	1,686	4,951	13,59	49,58	145,2	240,8	N/A
Initial water PAH content, µg/L	0,000	0,000	0,000	0,000	0,000	0,000	0,000
Post-adsorption water PAH content, calculated, µg/L	0,02629	0,07624	0,2039	0,7207	2,103	4,752	0,000
Initial AC PAH content, µg/g	0,000	0,000	0,000	0,000	0,000	N/A	0,000
Post-adsorption AC PAH content, measured, µg/g	0,09034	0,1726	0,6962	2,657	41,47	N/A	0,000
System PAH losses	0,28600	0,5790	1,9985	5,753	21,75	-9,32	0,000

Table 4.1: PAH contents of the O-rings, water, and active carbon, before and after the adsorption experiment.

Please note again that the rings measured for the initial ring content and the post-adsorption rings are not identical; the increase in content is not for the same ring. Not applicable refers to the control sets that did not contain charged rings (set 6) or carbon (set 7). Zero values for initial water and carbon content were based on provided properties of the materials, post-adsorption carbon PAH content including zero values for control set 7 were obtained by measurements.

Distribution coefficients

With the above data, methanol:silicone and silicone:water distribution coefficients could be calculated. Please note that due to oversights mentioned before, the calculated silicone:water distribution coefficients are expected to be inaccurate.

	FLU	PHE	FUR	PYR
Methanol : silicone K, measured	3,6	4,85	4,85	5,05
Methanol : silicone K, literature (L/L)	2,75	3,68	3,97	3,53
Silicone : water log K, measured	4,61	4,53	4,95	4,92
Silicone : water log K, literature (L/L)	3,45	3,52		

Table 4.2: distribution coefficients for the PAH exchanges in the experiment: methanol : silicone, silicone : water, and water : active carbon. Literature values are from Smith et al., 2010.

Log K values vary up to 1,5 L/L from literature values. Silicone:water K values for fluoranthene and pyrene could not be found and an average of the K values for fluorene and phenanthrene has been used for water concentration calculations; since PAH contents were extracted only for fluorene and phenanthrene, this did not affect the parameter calculations presented below.

Isotherms

With the data from the mass balance, water and adsorbed concentrations of PAHs were calculated in µg/L and µg/kg and plotted as adsorption isotherms for the two PAHs that had been extracted from the carbon, fluorene and phenanthrene. The isotherms can be seen in figures 4.4 and 4.5.

Adsorption of fluorene to GAC830W

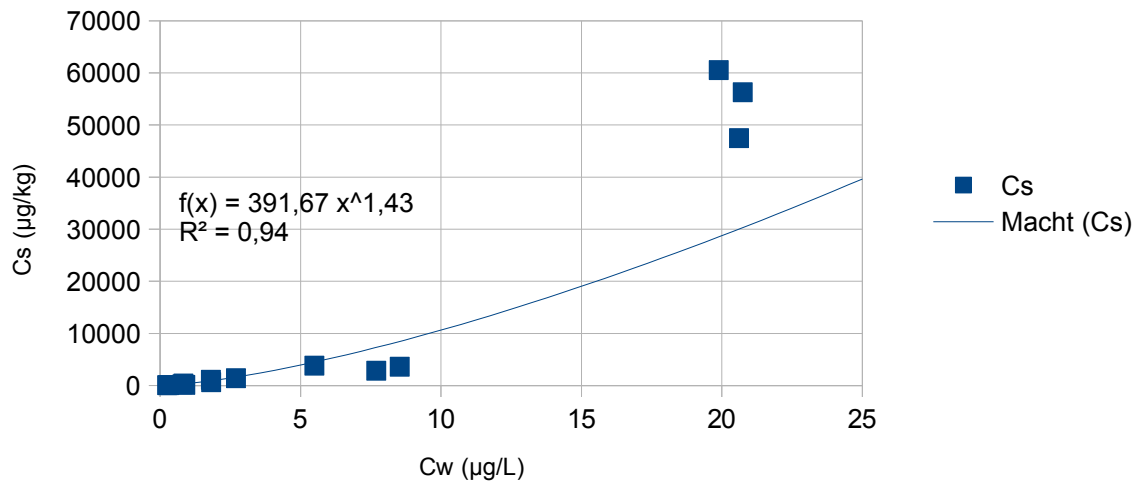


Figure 4.4: adsorption isotherm of fluorene to GAC830W activated carbon

Adsorption isotherm of phenanthrene to GAC830W

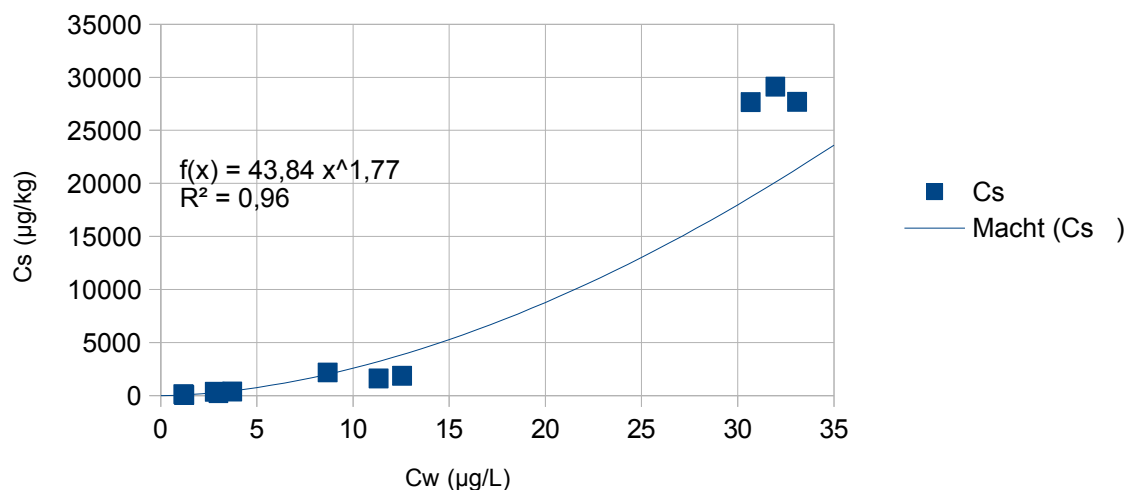


Figure 4.5: adsorption isotherm of phenanthrene to GAC830W activated carbon

It can be seen that these isotherms do not fit the shape of a Freundlich curve. Fitting these isotherms with a power function similar to the Freundlich equation generates the best fit, or highest R^2 . The name Freundlich will be used to describe this function despite the exponent being >1 .

Adsorption parameters

The values for the distribution coefficient and the exponent in table 4.3 describe the isotherm:

	Fluorene	Phenanthrene
Freundlich distribution coefficients, calculated (log K)	2,59	1,64
Freundlich distribution coefficient, literature (log K)	2,83	2,44
Freundlich exponent, calculated (1/n)	1,43	1,77
Freundlich exponent, literature (1/n)	0,604	0,406

Table 4.3: adsorption parameters, calculated and literature values. Literature values from Walters and Luthy, 1984.

The parameters differ from literature values $\log K$ for fluorene is closest to literature values but the found $\log K$ for phenanthrene is much (0,80 $\log K$) lower. In the case of the exponent the difference is very large, notably exceeding 1 while a Freundlich exponent is typically <1 .

5. Discussion

Control sample set 6 variations

The charged rings show an expected increase in PAH content and relatively consistent associated methanol:silicone distribution coefficient.

The post-adsorption rings show a marked decrease in phenanthrene and fluorene content, while fluoranthene and pyrene remain at a similar level. This could be due to a higher adsorption capacity of the black carbon for the two first mentioned PAHs.

The high variation in post-adsorption ring PAH contents in sample set 6, compared to the lower variation in sets 1-5 are remarkable, especially since the rings for sets 5 and 6 were both taken from charging set 5. Certain rings from sample set 6 showed increased PAH contents as compared to charging set 5, and some showed contents near identical to those found in sample set 5, suggesting either unknown processes occurring in certain sample erlenmeyers, measurement inaccuracies which are however unlikely since all other measurements provided consistent results, or some unknown failure in the extraction and analysis of the PAH contents of the rings from sample set 6. A repeated study of PAH content changes in charged silicone O-rings in water over several days could clarify what happened.

System losses

The decrease in fluorene and phenanthrene contents between the charged and post-adsorption rings is also seen in the amounts of PAHs extracted from the carbon: only the mentioned two PAHs have been extracted and in similar proportions.

However, in total quantity a large loss was noted between PAHs originally added to the erlenmeyer and the PAHs extracted from the rings, the carbon, and calculated to be in the water.

Low losses in sample set 6 suggest them to be related to carbon adsorption; maybe the PAHs adsorbed to the carbon too strongly for the pentane extraction to remove for analysis. According to Jonker, 2002, different solvents differ widely in their efficiency in extracting PAHs from black carbon – amounts extracted differed up to 95% from the highest measured extraction for certain solvents; pentane was however not studied. Inefficient extraction by pentane could account for the losses observed. Other possibilities are high water contents, which is however not consistent with the silicone:water distribution coefficient from Smith et al. and indications from the analysis of the water from control set 6 showing very low water concentrations; photodegradation, which is however not likely due to UV-filters installed on laboratory windows; or measurement errors of over 50% of total ring PAH contents, inconsistent with the low variation in measurements of i.a. the charged rings. Inefficient carbon extraction is most likely the cause of the losses in PAH contents.

Adsorption parameters

The found distribution coefficients for silicone to methanol are different from the values found in literature. This could be due to a difference in materials used, or in accuracy in the extraction and analysis of the contents of the rings. The silicone to water coefficients found differ widely from values found in literature, caused most likely by the mentioned oversights in analysis.

With the current results, the Freundlich distribution coefficients are somewhat similar to literature values, with variation from 0,4 to 1 $\log K$; the exponents differ more at 0,8 and 1,3 $1/n$, being well above 1, resulting from a concave isotherm.

Hypothetical isotherms

The isotherms found do not fit the shape of a Freundlich curve, which can be due to the incomplete recovery of PAHs from the different phases. Below are two hypothetical adsorption isotherms constructed by adding the PAH losses to the recovered PAHs in the calculation of C_s .

Hypothetical adsorption isotherm of fluorene to GAC830W

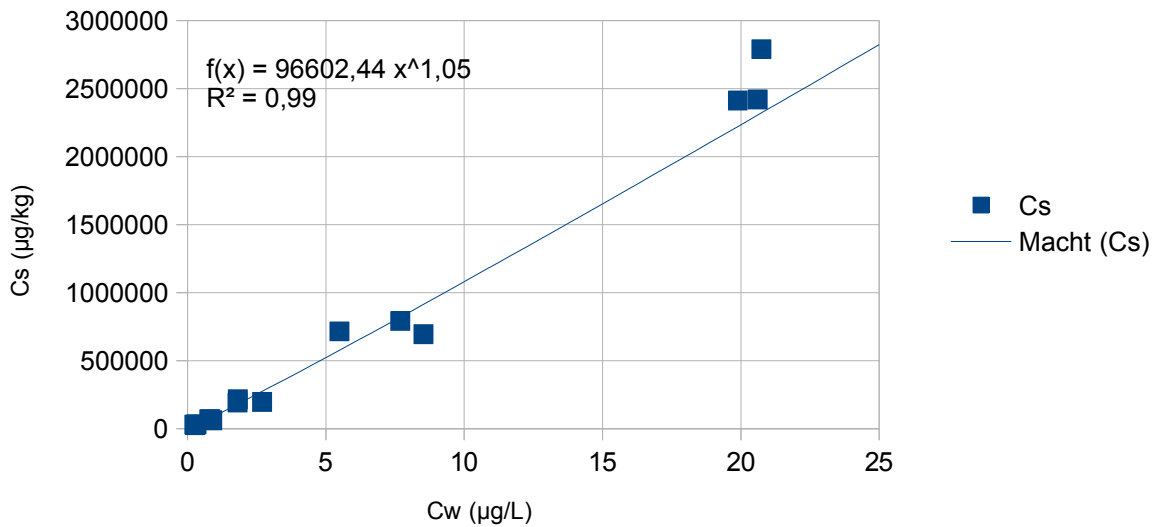


Figure 5.1: hypothetical adsorption isotherm of fluorene to GAC830W activated carbon

Hypothetical adsorption isotherm of phenanthrene to GAC830W

'Adsorbed' concentrations calculated from dosing ring losses in PAH content

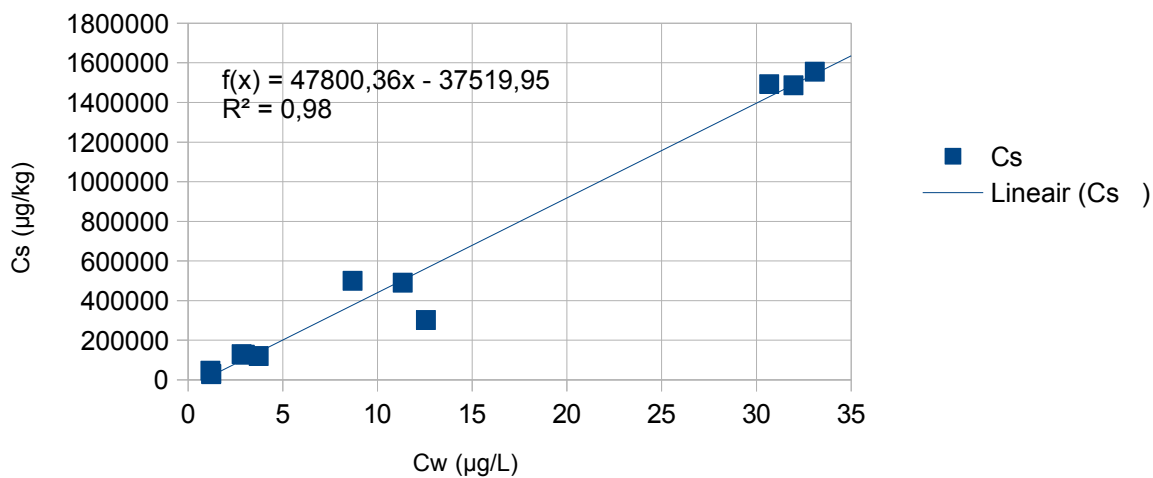


Figure 5.2: hypothetical adsorption isotherm of phenanthrene to GAC830W activated carbon

It can be seen that the adsorption isotherm is still concave and not of the general convex shape of either the Freundlich or Langmuir isotherm. It does however resemble a linear isotherm, and in the case of phenanthrene a linear curve provided the best fit. A log K would in this case be 4,98 for fluorene, and 4,68 for phenanthrene, but the Freundlich exponents would still be very different from literature values. The limited range of 1 to 35 µg/L for water concentrations might be too limited to see the curve in the isotherm, in which case a Freundlich exponent could still exhibit with a larger range of dosing concentrations. However this is hypothetical since the PAH losses cannot be conclusively be linked to inefficient carbon extraction.

6. Conclusion

To answer the research question with a degree of certainty, more research is required.

The parameters for the adsorption of fluorene, phenanthrene, fluoranthene and pyrene to Norit GAC830W have been found. They are however very different from literature values due to lack in laboratory proficiency (oversights), specific materials and maybe inefficient methods used: the efficiency of pentane extraction for PAH recovery from active carbon should be assessed to improve the reliability of the results. It is likely that only a small portion has been extracted with this method with far reaching impacts for the results and parameters found.

The method of passive dosing with silicone O-rings seems to be useful but can only be properly evaluated when accurate results have been achieved; furthermore the high variation in PAH contents from the O-rings in a control sample set raises questions about the behaviour of the rings in water during long shaking periods (7 days). Before any assessment of the usability of the O-rings in adsorption experiments such as this one can be given this variation should be explained.

The near linear shape in the hypothesized adsorption isotherm for GAC830W for PAH concentrations from 1 to 35 $\mu\text{g/L}$ is noticeable. With greater range in dosing concentrations and more efficient methods, maybe a Freundlich or other convex adsorption isotherm can be found, but this can not yet be verified.

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